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MILITARY SPECIFICATION

UNIT, DENTAL OPERATING AND TREATMENT OVER PATIENT TYPE, CHAIR MOUNTED

This specification is approved for use by the Defense Personnel Support Center based upon currently, available technical information but it has not been approved for promulgation as a revision of MIL-D-42007. It is subject to modification. However, pending its promulgation as a coordinated military specification, it may be used in acquisition.

1. SCOPE

1.1 Scope. This specification covers an over-the-patient, chair side mounted dental delivery unit.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-M-7298 - Manual, Technical, Commercial Equipment.

AMSC N/A FSC 6520

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Personnel Support Center, Directorate of Medical Materiel, DPSC-RST, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter

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STANDARDS

FEDERAL

FED-STD-H28	-	Screw-Thread Standards for Federal Services.

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.
MIL-STD-810	-	Environmental Test Methods.
MIL-STD-889	-	Dissimilar Metals

(Copies of specifications and standards required by contractors in connection with specific acquistion functions should be obtained from the contracting activity or as directed by the contracting activity).

2.2 Non-Government publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the non-Government documents which are current on the date of the solicitation.

AMERICAN NATIONAL STANDARDS INSTITUTE/AMERICAN DENTAL ASSOCIATION

ANSI/ADA No. 47 Specification for Dental Units.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, New York 10018).

UNDERWRITERS' LABORATORY, INC. (UL)

UL 544 Standard for Safety Medical and Dental Equipment.

(Application for copies should be addressed to the Underwriters' Laboratories, Inc., 207 East Ohio Street, Chicago, IL 60611, 333 Pfingsten Road, Northbrook, IL 60062, 1285 Walt Whitman Road, Melville, LI, NY 11746, and 1655 Scott Blvd., Santa Clara, CA 95050).

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification takes precedence. Nothing in this specification, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Materials. Materials shall be as specified herein. When materials are not specifically designated, those used shall be new, be of the same or better quality as used in commercial practice, and entirely suitable for the purpose intended. When dissimilar metals are used in contact with each other, suitable protection against galvanic corrosion shall be applied in accordance with MIL-STD-889.

3.2 General requirements. The delivery unit shall comply with the applicable requirements of American National Standards Institute/American Dental Association (ANSI/ADA) Specification No. 47 for dental operating units, Underwriters' Laboratories, Inc. (UL) 544 Standard for Safety Medical and Dental Equipment, and additional requirements and exceptions as specified herein.

3.2.1 First article. When specified in the contract or purchase order, a sample shall be subjected to the first article inspection (see 4.3 and 6.2).

3.3 Standard commercial product. The dental unit shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, shall be included in the dental unit being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements, or manufacturer's catalogs or brochures, and represents the latest production model.

3.4 Description. Shall be an over-the-patient, chair side mounted dental unit which shall provide unrestricted, efficient and safe delivery of unit related instrumentation in the performance of sit-down and stand-up dentistry. The unit shall consist of a mounting bracket compatible with the dental patient chair, a support post, a control head support arm assembly, a control head, a bracket table, a foot control, a cuspidor with drinking water cup fill, a utility junction box or boxes, and certain unit related instrumentation specified herein. The unit shall be easily and quickly converted into functionally equivalent right-and left-handed configuration either without tools or with tools supplied by the manufacturer. If tools are furnished by the manufacturer, they shall be accessibly stored within the unit.

3.5 Design, construction and performance.

3.5.1 Mounting bracket/support post. A heavy duty metal swing mounting bracket shall be furnished and shall be capable of being attached to the structural member of chair to permit right/left hand conversion.

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The right/left hand conversion shall be capable of being accomplished within 5 minutes when tested as specified in 4.4.3.1. A 2 inch diameter vertical steel support post capable of being mounted to the mounting bracket shall be furnished. A duplex receptacle shall be provided on the unit support post, when specified and shall conform to the system voltage selected. (see 6.2.1).

3.5.2 Control head support arm assembly. The control head support arm assembly shall consist of a rigid arm and an adjustable arm. One end of the rigid arm shall mount on the unit support post, and the other shall connect to an adjustable arm which shall support the control head. The control head support arm assembly shall be fully adjustable and movable so as to provide transthoracic instrument delivery. The control head support arm assembly shall be fully adjustable and movable so as to provide over the patient instrument delivery. The maximum reach of the control head shall be to a position where the control head is located lateral to the chair back, on the side opposite the pole mount, with the handpiece holders even with the top of the chair back, with the chair back in the horizontal position. The arm assembly shall not infringe on the patient when the control head is in this position. The total length of the arm assembly shall not be longer than 66" when measured along the arms from the center of the pivot at the attachment to the unit to the point of attachment of the control head. The adjustable arm shall pivot around the free end of the rigid arm in a horizontal plane for a minimum of 90 degrees when tested as specified in 4.4.3.3.

The arm shall be capable of supporting a 10 pound load on the bracket table without deflecting more than 1/4 inch when tested as specified in 4.4.3.2. The support arm shall be equipped with an adjustable counter balance spring. All pivot points, bushing, and bearings shall be self-lubricating. The control head shall remain stable and level throughout its range of motion. The control head and its attachments shall move vertically with the chair so as to minimize the need for height adjustment. The arm shall be furnished with a positive locking device at each selected position and shall be equipped with positive stops at its extreme limits of movement so as to prevent collision with walls, light post, or its own segments and to prevent damage to its internal components. The control head support arm shall provide a means of easy and unrestricted access to its internal components and contents.

3.5.3. Control head system. The control head enclosure shall be fabricated from a high impact plastic or corrosion resistant metal. The control head shall have an easily removeable cover or panel to allow unrestricted access for repair or service. Dimensions of control head excluding the bracket table shall not exceed 6 inches in height, 12 inches in depth, and 20 inches in width.

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Shall be capable of controlling a minimum of three automatically activated handpieces (see 6.2.1). Shall include a 3-way air-water syringe with a removeable, sterilizable tip and controls for air and water flow adjustment. The following controls shall be provided: master on/off control with a visible indicator, air pressure gauge, coolant water on/off, drive air adjustment for each handpiece, and a water and coolant air adjustment for each handpiece. All controls shall be easily accessible and operable and shall be clearly labelled. All tubing shall be color coded and shall be securely attached to barbs or fittings by positive clamps. Clamps shall be capable of being easily installed and removed. Barbs and fittings shall be constructed of durable non-corrosive materials and shall be individually replaceable.

3.5.3.1 Air control. A master on/off switch, with a visible indicator, shall be provided to activate all required valves and switches for the operation of the system. The switch shall be located on the front panel of the control head. Drive air pressure adjustment valves shall be provided for individual air pressure adjustment.

3.5.3.2 Handpiece control block. Each handpiece shall have a control block or a control block manifold sequentially mounted to a main control block. The control block mechanism shall be equipped with valves which are positive in action and capable of sustained operation in the presence of particulates smaller than 5 microns. The air and water supply connections to the control block shall be assembled with necessary gaskets and hardware to provide for secure instrumentation.

3.5.3.3 Water coolant control. Control head shall have easily accessible external controls to allow coolant water adjustment for each handpiece and a manual coolant water on/off control. The water control shall also be controlled by a relay operated by signal from the foot control.

3.5.3.4 Water anti-retraction. To prevent water line contamination, the unit shall not aspirate fluids back into instrument tubing when tested as specified in 4.4.3.4.

3.5.3.5 Pressure indicator. An air pressure gauge or bar graph shall be mounted on the front panel and shall indicate the drive air pressure. The indicator shall have a range of at least 0 to 60 pounds per square inch.

3.5.3.6 Instrument holders. Each handpiece shall be securely retained in a readily accessible handpiece holder which shall automatically activate the handpiece upon its removal. The automatic holders shall contain a self-cancelling feature to allow only one handpiece to operate at a time. All handpiece holders shall be of the same size and design.

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The holders shall be designed to securely hold all instruments. As a user specified alternate, the handpieces and air/water syringe shall be suspended above the control head by means of counterbalanced, folding levers (see 6.2.1). The holders shall be constructed of a durable plastic and shall be attached to the handpiece holder bar/positioning handle. The holders shall be equipped with a manual lockout control to allow service or bur change. The holders shall be mounted in such a position so as not to interfere with easy access to the tray. As a minimum, each holder shall be adjustable about the holder bar from a vertical position through a 90 degree range and shall be adjustable in its lateral position.

3.5.3.7 Three-way syringe. The unit shall include a three-way syringe capable of producing a fine stream of air, water, or a mist of air and water and shall be provided with adjustments for flow control. Shall consist of a tip, syringe head and handle. The syringe tip shall be corrosion resistant steel and shall be autoclavable. The tip shall be rotatable for directional adjustment without loosening the attachment mechanism. The syringe tip shall have an O ring for positive air-water separation and to prevent leakage. The tip shall be easily removeable to allow for autoclaving. The head shall be made of stainless steel, chrome plated brass, or anodized aluminum. The water and air push buttons shall be easy to remove for valve maintenance by use of a simple tool, such as a screwdriver. The handle shall be made of anodized aluminum, stainless steel, or chrome plated brass.

3.5.4 Bracket table. A bracket table shall be attached to control head or support arm and shall be designed to allow convenient service access to control head for maintenance. The bracket table dimensions shall be large enough to accommodate standard bracket table paper (8 inches by 12-1/4 inches with rounded corners).

3.5.5 Foot control. Shall be a disc type foot control having a base fitted with anti-skid pads, and a scratch resistant cover designed to exclude dirt and debris. Shall be capable of varying handpiece speed in increments of 5 psi or less to permit easy operation and accurate control when tested as specified in 4.4.3.5. The foot control shall include an on/off switch for coolant water. Foot controller hose shall be uncoiled and at least 5 feet in length.

3.5.6 Cuspidor. The cuspidor shall be mounted on the vertical post and shall move (raise and lower) with the chair. The housing shall have an easily removeable cover to allow access for service. Shall feature an 8 or 9 inch opaque glass bowl with a drinking water dispenser. The bowl shall be easy to clean, be insusceptible to water spotting, and shall be capable of rinsing and flushing.

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The cuspidor and drinking water dispenser shall be equipped with a means of adjusting the volume of flow of water from 0 to 30 gallons per hour (GPH) at 30 psig. The bowl rinse water shall cover the entire bowl to carry debris down the drain. The bowl shall show no dry spots when tested as specified in 4.4.3.6. Drainage shall be vacuum or gravity assisted as specified (see 6.2.1). A 1/4 inch female quick disconnect water outlet with both check and control valves all fabricated of corrosion resistance metal shall be provided for use with accessory equipment. Sufficient drain tubing shall be furnished in the umbilical hose for connection to the utility junction box as applicable.

A positive seal around the bowl shall be provided to secure the bowl in its socket and reduce the possibility of leakage. The cuspidor assembly shall have a 50 mesh, non-corrosive screen type, solid trap for collection of solids. The cuspidor shall be equipped with positive stops at its extreme limits of movement so as to prevent collision with the dental chair. The cuspidor shall be equipped with momentary time delay controls when specified (see 6.2.1).

3.5.7 Assistant's instrumentation. Assistant's instrumentation shall be provided when specified and shall consist of: a high volume evacuator (HVE) handpiece with a rotary control valve, a saliva ejector with a rotary control valve, hoses, a 3-way air/water syringe, and suitable holders shall be provided, when specified (see 6.2.1). A rotary control valve (vacuum shut-down valve) shall be provided between the utility and the collection canister so as to permit isolated shut-down of the vacuum utility for routine maintenance service, when specified (see 6.2.1). Shall be mounted on a telescopic or scissor type arm and shall include a vacuum solids collection canister with a removeable 20 mesh non-corrosive screen. Location and movement of the arm shall allow unrestricted efficient access to the patient and all required instrumentation. Shall have a left-hand or right-hand installation capability.

3.5.7.1 Air venturi vacuum system. An air venturi evacuation system consisting of an air venturi vacuum device, a high volume evacuator (HVE) with a control valve, a saliva ejector with a rotary control valve, hoses, a 3-way air/water syringe, and suitable holders shall be provided, when specified (see 6.2.1). Instrumentation shall be mounted on a telescopic or scissor type arm.

The HVE shall be capable of a flow of no less than 2 cubic feet of air per minute at a negative pressure no less than 4 inches of mercury. The saliva ejector shall be capable of a flow of no less than 1 cubic foot of air per minute at a negative pressure of 15 inches of mercury. This performance shall be exacted at 50 psi and 2 cfm of air.

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3.5.8 Junction box. The junction box or boxes, as applicable, shall be an interface for connecting the dental delivery system, to the building utilities, (air, water, electricity, vacuum, and drain). The box shall be equipped with switches and valves to permit isolated shut-down from all utilities for maintenance or repair of the unit. The junction box shall have maximum dimensions of 15 inches in width, 10 inches in depth and 5 inches in height and be constructed of a heavy gauge corrosion resistant metal. The outside surface of the box shall have a smooth finish and shall be easy to clean. The junction box shall be equipped with a lid which shall be quickly and easily removeable without tools so as to allow quick and unrestricted access. The box shall be capable for flush or raised mounting and have slotted mounting holes and shall be furnished with all fastening devices required to mount the case to concrete, wood, paneling or sheet rock (see 6.2.1). The utility junction box installation shall not require building utility stub-ups (air, water, vacuum, drain, electrical) to extend into the confines of the box, and shall not require stub-ups to conform to any template configuration other than a requirement that all stub-up ends or connection fittings be not more than 18 inches from the center of the utility box location. All hardware, tubing and wiring connection of the stup-ups to the junction box shall be provided by the unit contractor.

The junction box shall have umbilical outlets with cover plates in each of its sides so as to permit neat and orderly installation. A single umbilical shall house all tubing and wiring between the unit and junction box and shall be provided with strain relief and protection from abrasion at both connections. All tubing or wires routed through holes shall be protected from abrasion with suitable grommets.

A duplex receptacle shall be equipped with junction box, when specified (see 6.2.1). An electrical receptacle shall be provided at the unit terminus of the umbilical to provide power to a light when a chair mounted light is required with system. The junction box shall be furnished with all necessary tubing and wiring to make it a functional, operating system with the dental unit, vacuum instrumentation, cuspidor and dental chair. The shut-off valves for compressed air or water shall be corrosion resistant metal.

A manual shut-off air valve, a filter and adjustable air regulator with a pressure gauge and moisture separator with a manual drain shall be provided. A valve designed to accomplish a remote on/off control of the air and water supply by a master toggle valve which is located on the control head shall be provided.

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A manual water shut-off valve, a filter and an adjustable regulator with a pressure gauge shall be provided. A water heater with an adjustable thermostat to allow adjustment of water temperature between 90° and 100°F at a minimum flow rate of 70 ml/min, shall provide warm water for use in the syringe. The electrical power to the heater shall be controlled by the unit master control switch.

The junction box shall be fitted with an easily replaceable cartridge or element type air and water filters capable of retaining particulates over 5 microns. The air and water filters shall be interchangeable.

3.5.9 Umbilical hose. The umbilical hose shall be a non-collapsible tubing as tested in 4.4.3.7 of sufficient length (at least 5 feet) to permit the raising, lowering, and rotation of the dental chair. The umbilical hose shall house all of the air, water, vacuum drain and electrical conduits tubes that are required to operate the dental delivery system. All compressed air and water tubing inside of the umbilical hose shall be a suitable plastic capable of operating safely at a pressure of 120 psi and temperature of 70 degrees F. The vacuum line shall be at least 7/16 inch ID tubing and shall not collapse under 20 inches of mercury vacuum. The umbilical hose and connections shall be designed to provide adequate protection for the internal tubing.

3.6 Sanitization. All exposed surfaces of the unit and control arm shall be smooth and made of materials which can withstand repeated wipe down with ADA acceptable disinfectants without deterioration of function or esthetics. No trim shall be attached such that its partial or complete loss will interfere with disinfection. All interface components that require frequent replacement such as the saliva ejector tubing, water tubing, holders on control head, etc., shall be capable of being santitized for the maintenance of proper infection control procedures. The cuspidor bowl shall be readily cleanable and shall not stain in the performance of dental procedures.

3.7 Electrical requirements. The electrical characteristics shall be as follows, as specified (see 6.2.1)

(a)	115 ± 10% volts,	60 ± 2 hertz
(b)	115 ± 10% volts,	50 ± 2 hertz
(c)	230 ± 10% volts,	50 ± 2 hertz
(d)	230 ± 10% volts,	60 ± 2 hertz

3.8 Finish. All surfaces shall be clean and free of foreign matter, flux, or welding residue. The finish of components shall be as specified, the finish shall be entirely suitable for the purpose intended. Finish on edges and surfaces shall be uniform and free from burrs, slivers, sharp edges, cracks, or other defects. All exposed surfaces shall be able to withstand repeated exposure to disinfectants without deterioration of function or esthetics. No trim shall be attached such that its partial or complete loss will interfere with disinfection.

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3.8.1 Plating and coatings. Shall be uniform in thickness and appearance; shall be free from blisters, peeling, visible porosity, or other defects. The base metal to be plated shall be free from flaws or defects which will be detrimental to the final finish and shall be subjected to such undercoating, polishing, cleaning, dipping, and plating procedures as are necessary to yield deposits with the desired function, quality, and appearance.

3.9 Workmanship. Workmanship shall be in accordance with best commercial practice. All components shall be clean, well finished, and free from defects which detract from their appearance or impair their serviceability.

3.9.1 Castings. Castings shall be sound and free from repairs, warping, or other defects that may render the casting unsound for use. Repair processes, such as welding, peening, plugging, or filling with cold solder or metallic paste shall not be permitted.

3.9.2 Welding, brazing and soldering. All welds, brazing, and soldering shall be durable and sound in every detail. Welding shall be in accordance with the latest practices recommended by the American Welding Society. Components shall be joined without any resultant distortion and shall be free from twists or strains.

All welds shall be produced by qualified welding operators with satisfactory welding equipment using satisfactory filler metal. Surfaces to be welded shall be free from grease, oil, oxides or any other foreign matter. Any cleaning method which will remove these substances and leave the surface clean is satisfactory. All aluminum parts shall be welded within 48 hours after a chemical cleaning treatment. Parts must be chemically recleaned if the 48 hour time period is exceeded. All weld joints shall blend into the adjacent metal in gradual, smooth curves. Beads shall be smooth and free of slag, excessive undercut, or excessive spatter. Sufficient metal shall be added to provide a suitable fillet or reinforcement unless otherwise specified. Welds shall not be oxidized. All welds shall be continuous with no visible voids, pins holes, or cracks.

Exposed welds shall be smooth and blend with adjacent surfaces. All welds shall be cleaned, discoloration removed, and be free from slag and spatter before finishing. Welds shall be ground, only to the degree necessary to remove sharp edges or projections which may cause defects to adjacent surfaces or can be injurious to the touch.

3.9.3 Machine work. Machining shall be in accordance with best commercial practice. All bolts, machine screws, and rivet holes shall be accurate and clean. The holes for screws and bolts shall be sized appropriately for the size screw or bolt used. All bolts, nuts, screws, and fasteners shall be completely fastened and tight.

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3.9.4 Metal fabrication. Metals (sheets, plates, and tubing) shall be free from dents, folds, or distortions. Material shall be straightened by methods that will not cause injury to the metal. Operations (such as bending, shearing, or cutting) shall be done neatly and accurately. Metal forming shall be made with controlled means to ensure uniformity of size and shape.

3.9.5 Hardware. All hardware shall be entirely suitable for the purpose intended and shall be completely compatible with the material to which it is attached or which it joins, including plated material. Any fastening device required to be loosened, tightened, or removed for repair, maintenance or adjustment shall not be swaged, peened, staked or otherwise deformed to prevent loosening of the device. All rotative fasteners shall be correctly torqued. All threaded parts shall comply with Federal Standard H28.

3.10 Identification marking. The unit shall have one or more nameplates, permanently fastened and accessible, containing at least the name of the manufacturer, type and model designation, serial number and electrical characteristics. The serial number shall be traceable by the manufacturer to the date of manufacture.

3.11 Tolerances. Where tolerances are not specified, standard commercial tolerances shall be permissible. Tolerances shall be held to a minimum where accuracy is of paramount importance for proper operation and interchangeability of component parts. Assemblies shall be precision-fitted for durable, smooth, and consistent functioning.

3.12 Interchangeability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable.

3.13 Service data. The contractor shall furnish, with each unit, two copies each of the technical manual which contains complete instructions for installation, operation, maintenance, and list of component parts. The manual shall comply with the requirements of MIL-M-7298. The installation manual and maintenance/repair manual shall include the following: step by step illustrated installation instructions; a list of special tools (if required); exploded drawings with part numbers; electrical and utility requirements; electrical and utility (compressed air, water, vacuum) schematic diagrams; pressure adjustment instructions and setting for air, water and vacuum; arm counterbalance adjustment; specific instructions to mount unit and swing mount adapter to the specified chair; checkout procedures to ensure all unit operations are functioning properly; telephone number for technical assistance if required; and a trouble shooting chart.

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The operations manual shall include the following: illustrated step by step procedures for the operation of the unit including the location and function of each control; step by step instructions for left to right conversion; pressure adjustments for air, water and vacuum; arm counterbalance tension adjustment; step by step instructions for adjustment of drive air, coolant water, chip air and anti-retraction for each handpiece; instructions for air and water adjustment for 3-way air-water syringe; instructions for the proper cleaning and disinfecting of all exposed surfaces; and a daily functional checklist.

As soon as practicable after award of the contract or purchase order, the contractor shall furnish to the Contracting Officer, via the cognizant Government inspector, two preliminary copies of his proposed technical manual, for review and comment by the Directorate of Medical Materiel, Defense Personnel Support Center. The preliminary manuals shall be submitted sufficiently early to permit adequate review by the Directorate of Medical Materiel, Defense Personnel Support Center (based on a maximum period of 30 days for such review) and to allow incorporation by the contractor of any required revisions and corrections in the final manual without delaying delivery under the contract or purchase order. In addition, the Contractor shall furnish 15 copies of the approved technical manual to the Contracting Officer for distribution. Should the supplier have furnished acceptable manuals for the identical item within two years preceding the date of the contract and propose to furnish manuals identical with those previously accepted, then the requirements for review of drafts and submission of 15 additional copies may be waived upon request to the Contracting Officer. Waiver of submission of 15 additional copies shall be at the option of the Government.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

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4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual and auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the materials, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the Item Identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements. Certificates of quality may be furnished for the following: material composition, hardness, tempers, torque requirements on springs, machining and casting requirements (such as flatness, concentricity, and finish), and plating or coating.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.3).

2. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is specified in the contract, the first article inspection shall consist of all examinations specified herein, and all testing specified herein and shall apply to the first article sample only (see 6.2). The failure of compliance with any examination or test shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Except as otherwise specified herein, sampling for inspection shall be performed in accordance with the provisions set forth in MIL-STD-105.

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4.4.1 End item examination.

4.4.1.1 Visual examination. End item examination shall be conducted in accordance with the following classification of defects. Examination shall include, but shall not be limited to, the defects listed in table I. The inspection level shall be II with an acceptable quality level (AQL) of 1.0 for major defects and an AQL of 10 for minor defects. The unit of product shall be the component of an end item or an end item, as applicable.

TABLE I. End item classification of defects.

	Major	Minor
Design and Construction:		
Design and construction of unit not as specified.	X	
Hardware:		
Components, such as pins, screws, and fasteners missing, broken, stripped, cracked or deformed.	X	
Fasteners loose.		X
Finish:		
Finish not as specified.	X	
Surfaces not clean, not free of foreign matter, flux, spatters, or other defects.		X
Damage or defects on exterior or interior surfaces, such as burrs, slivers, cracks, or sharp edges which effect function or serviceability.	X	
Damage or defects on exterior or interior surfaces, such as burrs, slivers, cracks, or sharp edges which do not effect function or serviceability.		X
Plating missing (affecting function).	X	
Plating missing (not affecting function).		X
Plating not uniform in appearance.		X
Plating not free from blisters, peeling, visible porosity, or other defects.	X	

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TABLE I. End item classification of defects (Continued)

	Major	Minor
Workmanship:		
Any component fractured, broken, punctured, torn, bowed, deteriorated, or malformed.	X	
Any component dented or creased.		X
Any component misplaced or not in proper alignment.	X	
Any component not connected or not joined properly.		X
Fastening device requiring loosening or removal is swaged, peened, staked, or otherwise permanently fastened.	X	
Components missing.	X	
Components do not fit or mate properly.	X	
Interface fits between components not proper (too loose, too tight/binding).	X	
Components not free from defects which may impair their serviceability.	X	
Components not free from defects which do not impair their serviceability.	X	
Removable components cannot be removed and replaced without difficulty.	X	
Components not properly assembled or aligned.	X	
Machine work:		
Coarse machine, tool, or die marks which affect function or serviceability.	X	
Coarse machine, tool, or die marks which do not affect function or serviceability.		X
Identification Marking:		
Identification marking missing, incomplete, not legible.		X
Identification marking incorrect or not permanent.		X
Marking not properly located.		X
Service Data:		
Service data not included.	X	

4.4.2 Dimensional examination. Dimensional examination shall be conducted for compliance with dimensions specified. Any dimension not within specification tolerance shall be a defect. The inspection level shall be S-3 with an AQL of 1.0. Unit of product shall be the component of an end item or an end item, as applicable.

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4.4.3 End item testing. End item testing shall be conducted in accordance with MIL-STD-105 and as indicated in table II, except that the acceptance number shall be zero for all sample sizes. Unit of product shall be one delivery unit.

TABLE II. Sampling for tests.

Component	Requirement	Test Procedure	Inspection level
Right/left hand conversion	3.5.1	4.4.3.1	S-2
Arm Locking mechanism loading	3.5.2	4.4.3.2	S-2
Adjustable arm horizontal angular rotation	3.5.2	4.4.3.3	S-2
Water anti-retraction	3.5.3.5	4.4.3.4	S-2
Foot control accuracy	3.5.5	4.4.3.5	S-2
Cuspidor bowl rinse	3.5.6	4.4.3.6	S-2
Umbilical hose	3.5.9	4.4.3.7	S-2

4.4.3.1 Right/left hand conversion test. The control arm shall be rotated from a right handed operator's position to a left handed operator's position.

4.4.3.2 Arm locking mechanism loading test. A weight of 10 pounds shall be placed on the bracket table. The unit shall be observed for deflection.

4.4.3.3 Adjustable arm horizontal angular rotation. The adjustable arm shall be rotated in the horizontal plane and the angular range of rotation shall be measured.

4.4.3.5 Water anti-retraction test. Shall be in accordance with ANSI/ADA No. 47, Specification for Dental Units.

4.4.3.5 Foot control accuracy test. Operate a handpiece at medium speed by means of foot control. Increase and decrease the speed in the lowest increment possible as indicated on the handpiece pressure indicator.

4.4.3.6 Cuspidor bowl rinse. Activate cuspidor bowl rinse and run for one minute. Observe bowl for dry spots.

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4.4.3.7 Umbilical hose. Apply a 200 pound force to hose with a plate (4 inches X 6 inches), with the hose crossing the short axis of the plate. There shall be no more than a 50% loss in the hose's minor diameter and the hose shall exhibit no memory. The hose shall not collapse.

5. PACKAGING

5.1 Preservation. Preservation shall be level C.

5.1.1 Level C. Each dental operating and treatment unit shall be packaged and cushioned in standard commercial containers of a suitable size and design that will afford adequate protection against deterioration and damage during shipment and storage.

5.2 Packing. Packing shall be level C.

5.2.1 Level C. Each packaged dental operating and treatment unit shall be packed in a substantial commercial exterior container (export when applicable) which will afford adequate protection to the contents during shipment from the supply source to the receiving activity. This pack shall conform to the carrier's rules and regulations.

5.3 Marking.

5.3.1 Interior packages. Interior packages shall be marked to identify contents and quantity.

5.3.2 Exterior container. Each exterior container shall be marked as specified in MIL-STD-129.

6. NOTES

6.1 Intended use. The delivery unit is intended to be used as an integrated system with chair for use in the treatment of dental patients.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number and date of this specification.
b. First article, when required, (see 3.2.1).
c. Options required, (see 6.2.1).

6.2.1 The following options are listed to allow the user to choose those items required to effectively treat patients at their facility. In each category, several options are given. If no option is checked, the first listed shall be the default value.

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a. Electrical requirements.

(1) 115 ± 10% volts, 60 ± 2 hertz
(2) 115 ± 10% volts, 50 ± 2 hertz
(3) 230 ± 10% volts, 50 ± 2 hertz
(4) 230 ± 10% volts. 60 ± 2 hertz

b. Number of handpiece holders.

(1) Three (3) required.
(2) Other; specify.

c. Assistant's instrumentation options.

(1) Instrumentation for use with central vacuum system.
(2) Instrumentation with air venturi vacuum system included.
(3) No assistant's instrumentation required.

d. Drain cuspidor options.

(1) Gravity drain cuspidor with drinking water dispenser and bowl rinse required.
(2) Vacuum drain cuspidor with drinking water dispenser and bowl rinse required.
(3) Momentary time delay controls required.

e. Utility junction box options.

(1) Flush mounted type.
(2) Raised mounted type.
(3) Duplex receptacle required on junction box.
(4) No receptable required on junction box.

f. Support post duplex receptacle options.

(1) Duplex receptacle required on support post.
(2) No receptacle required on support post.

g. Vacuum shutdown valve options:

(1) Vacuum shutdown valve required.
(2) No vacuum shutdown valve required.

h. Instrument delivery.

(1) Mounted on horizontal bar attached to front of the control head.
(2) Mounted on folding levers above the control head.

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6.3 Installation requirements. In order to ensure proper compatibility of the utility junction box and other components of the central unit with the specific facility, the following information must be furnished by the facility.

(a) Compressed air source: (1) compressed air line diameter (ID and OD) and material of construction (PVC, copper etc.); connector type at utility box.

(b) Water supply: (1) water line diameter (ID and OD) and material of construction (copper, flexible tubing etc.); (2) connector type at utility box.

(c) Vacuum connections: (1) vacuum line diameter (ID and OD) and material of construction (copper, flexible tubing etc.); (2) connector type at utility box.

(d) Drain connections: Drain pipe diameter and location.

(e) Location requirement for junction box and dimensions of utility box.

(f) Tubing requirements: (1) handpiece connector type (please list names of handpieces, fiber optic systems and indicate specific connector types required). If more than one type is required for each control unit, please indicate the number of each required for one control head; (2) tubing type desired (coiled or straight); (3) length of tubing required.

MILITARY INTERESTS:

Custodians:

Army-MD
Navy-MS
Air Force - 03

PREPARING ACTIVITY:

DoD-MB

Agent:

DLA-DM

Project No. 6520-1961

Location: NCR Disk #3

Civil Agency Coordinating Activities:
VA-OSS
PHS

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